

Classification of living things

Have you ever tried to observe and classify the living things in your surroundings? What are the main differences among these living things? Have you ever thought how many living things exist on earth? And how are they classified into plants and animals?



Figure 1.1: Coral reef

Does this creature look like a plant or an animal?
Is it a plant or an animal?
Can you explain your answer?



Figure 1.2: Mushrooms

"I am neither classified under plants nor animals."
WHY?

In this chapter, you will learn about:

- Introduction to the main kingdoms (Bacteria, Algae, Fungi, Plants and Animals)
- Classification of the kingdoms
- Classification and characteristics of animals (vertebrates and invertebrates)
- Classification and characteristics of plants (flowering and non flowering plants)
- Classification of flowering plants (monocotyledonous and dicotyledonous plants)
- Characteristics of monocot and dicot plants (leaf shape, venation, seed and number of floral leaves)

All the students will be able to:

- ✓ Define classification.
- ✓ Explain the need and importance of classification.
- ✓ Differentiate between vertebrates and invertebrates according to key characteristics.
- ✓ Identify vertebrates and invertebrates from their surroundings.
- ✓ Classify vertebrates into mammals, reptiles, fish, birds and amphibians on the basis of their characteristics.
- ✓ Identify key characteristics of worms and insects.
- ✓ Compare flowering and non-flowering plants.
- ✓ Classify the flowering plants into two major groups and give examples of each group.
- ✓ Compare the structure of monocot and a dicot seed.
- ✓ Compare the structure of monocot and a dicot leaf in terms of its shape and venation.
- ✓ Differentiate between the structure of monocot and dicot flower in terms of number of floral leaves.

Introduction to the main kingdoms

- 1 Define classification.
- 1 Explain the need and importance of classification.



Think about it!

Which features can help you in classifying living things?



Figure 1.3: Animal kingdom

In the previous class, you have studied the needs, characteristics and life cycle of living things. You have also studied that living things can be classified on the basis of the food they eat. Now we will study another way to put living things in groups. Observe all the living things in your surroundings; a scientist calls these living things, **organisms**. Many living things are alike in some way. They can be put into groups according to the ways they are alike. Putting living things into groups is called **classifying**. In order to study living things easily and systematically, we need to classify them.

Unscramble the given word:
REVETARBTE

Hint: The animals that have backbone.

For a long time, living things have been classified into the plant kingdom and the animal kingdom, according to their common characteristics. With the advancement of technology, scientists have observed more details in living things. As a result, living things are now classified into five different kingdoms.

Do you know that living organisms are made up of cells? Cell is the basic unit of living things. Cells are also microscopic. Most cells have nucleus and cell wall.



Figure 1.4: Bacteria

Bacteria

Unicellular organisms (single-celled), have cell wall but do not have proper nucleus.

They occur in air, water, soil or inside other organisms. Many of them cause diseases. But some are used to make medicines, yogurt, and cheese.

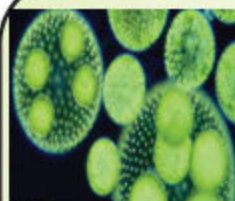


Figure 1.5: Algae

Algae

Mostly unicellular with proper cell wall and nucleus.

They contain chlorophyll and make their own food just like plants by photosynthesis. They live mainly in water. Many of them are used as food. Some cause diseases; for example, Cutleria, Ulva, and Volvox.



Figure 1.6: Fungi

Fungi

They are mostly multicellular, with modified cell wall and nucleus.

They do not have chlorophyll and do not photosynthesize. Instead, they feed on material like human foods and dead plants or animals. Many of them cause diseases. Some are used to make medicines and bread.

Living things



Figure 1.7: Plants

Plants

Multicellular organisms (many-celled) with modified cell wall and nucleus. Also contain the green substance chlorophyll and make their own food by photosynthesis. They have proper roots, stems, leaves, flowers and also produce fruits.



Figure 1.8: Animals

Animals

They are multicellular organisms with modified nucleus but do not have cell wall. They cannot make their own food; feed on other organisms and usually move around. They are classified into vertebrates and invertebrates.

Activity1: Your garden, farm or nearby park can be a living place for many living things. Make a list of all the living things that visit or live in , any one of these places.

What I need:

- Observe in your garden.
- Observe how many different types of animals and plants visit or live in your garden.
- Talk to your family members, elders, teachers and senior fellows.
- Use hand lens if required.

What to do:

1. Your teacher will form groups of three students in a team.
2. Ask each member to observe and record physical characteristics of any one living thing from the garden (like bird, earthworm, butterfly, cat, grass, flower, etc.)
3. Draw or bring a picture of the living thing.
4. Along with the teacher, observe the living thing and its physical characteristics.
5. Now share the name/picture of the living thing, along with the characteristics, with group members.

What I observed:

Living things from the garden	Physical characteristics of any one common living thing

Activity questions:

1. What characteristics were most common in all living things?
2. Can you classify them according to their physical characteristics?

Teacher Note: Arrange and provide the materials for classification and facilitate students by observing the living things. Also ask the students questions during observation. Also help students in writing at least three characteristics of living things.

Classification and characteristics of animals

 Differentiate between Vertebrates and Invertebrates according to key characteristics.

Vertebrates and invertebrates:

All the animals on the Earth form a kingdom called the **animal kingdom**. Some are microscopic; so small that they can only be seen under a microscope. Others are gigantic, such as the enormous blue whale.

Which part of your body helps you stand tall or sit up straight? It is your backbone. You are a member of a large group of animals that all have backbones. Animals with backbones are called **vertebrates**. Fish, amphibians, reptiles, birds and mammals are all vertebrates. Some animals do not have backbones. These animals are called **invertebrates**. Worms, spider and insects are all invertebrates.

Activity 2: Find out the names of different animals.

Find the names of four vertebrates and four invertebrates hidden in the word search. Then write them in the correct group.

B	R	A	B	B	I	T	B	U	D
E	A	G	I	R	A	F	F	E	L
E	W	O	F	H	E	P	R	U	W
T	O	G	K	L	C	M	O	T	H
L	R	N	F	R	Y	S	G	I	A
E	M	L	I	O	N	O	J	E	L
R	S	P	I	D	E	R	M	R	E



Vertebrates

1. _____.
2. _____.
3. _____.
4. _____.

Invertebrates

1. _____.
2. _____.
3. _____.
4. _____.

Teacher Note: Engage students in the activity and help them write the names of four vertebrates and four invertebrates.

Animals know how!

The word "animal" comes from a Latin word meaning soul or breathe.

There are more than 30,000 kinds of fishes.



There are over 9,000 kinds of birds.



There are over 8,00,000 kinds of insects.



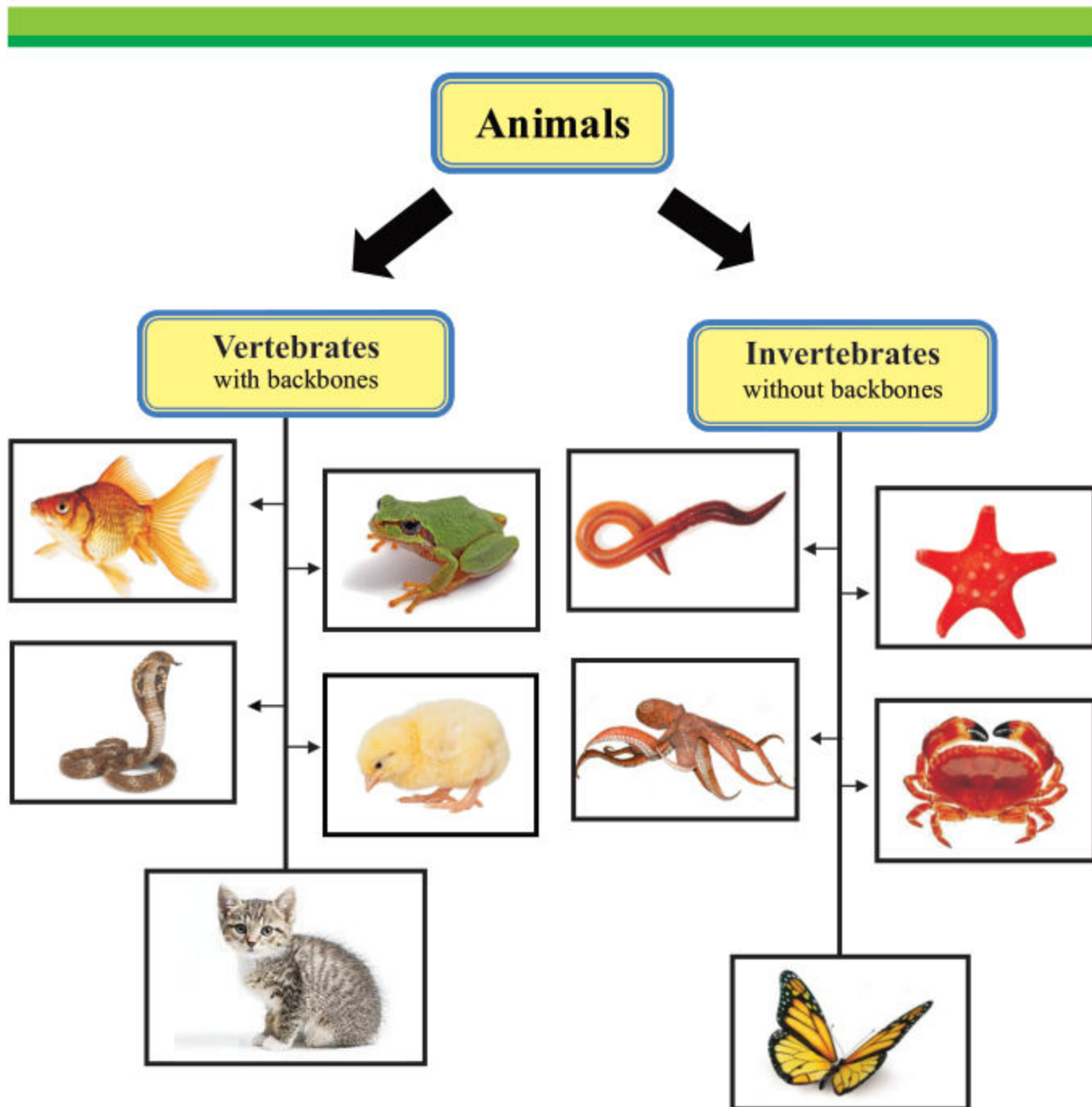


Figure 1.9: Classification of Animal (Vertebrates and Invertebrates)

Did you notice that the organisms are classified through divisions into two smaller groups each time? This is called a **Dichotomous key**.



Unscramble the given word:

LOYZOGO

Hint: The scientific study of animals.

Activity 3: With your parents/teacher, plan a visit to a zoo, or observe animals on a farm. Observe how many classes of animals are found there. What environment is provided to each class of animal at the zoo? Do not forget to take a notebook with you to record all these things.

Identify vertebrates and invertebrates from their surroundings.

What I need:

- Observe the animals.
- How many types of animals are found there?
- Observe differences among them according to their physical characteristics and way of living
- Talk to your family members, elders, teachers, class fellows and senior fellows.
- Also visit a library, visit WWF website or WWF center in Karachi.
- Watch a National Geographic program.
- Use hand lens if required.



Figure1.10: Children are observing the animals

What to do:

1. Your teacher will form groups of four or five students in a team.
2. Ask each member to observe and record physical characteristics of one vertebrate and one invertebrate from the zoo. Also observe the living place of each animal.
3. Draw or bring a picture of the vertebrate or invertebrate.
4. Observe the vertebrates and invertebrates animals with your teacher and record their physical characteristics.
5. Now share the name/picture of the vertebrates and invertebrates, along with their characteristics, with group members.

Teacher Note: Discuss with students and assist them in identifying and recording the characteristics of vertebrates and invertebrates. The teacher should also bring pictures of vertebrates and invertebrates for observation.

What I observed: (Remember to write your team member's names.)

Team members	Name/Picture of vertebrates and invertebrates	Characteristic of vertebrates and invertebrates	Way of living

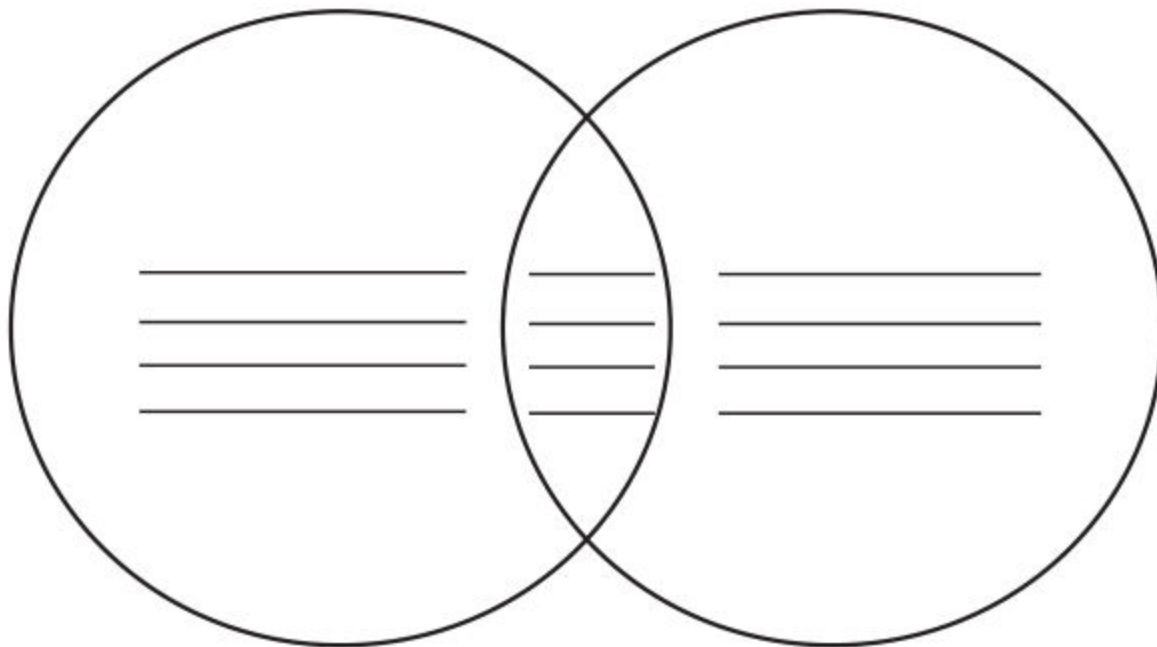
Activity questions:

1. Which team member shared the most interesting characteristics? What were the most interesting and unique characteristics?
2. What characteristics were most common in vertebrates and invertebrates?
3. Compare vertebrates and invertebrates? How are the characteristics of the vertebrates and invertebrates similar and different? Write their similarities and differences.

Vertebrates

Similarities

Invertebrates



Teacher Note: Engage students in the activity and help them identify similarities and differences, in vertebrates and invertebrates.

Vertebrates

 Classify vertebrates into mammals, reptiles, fish, birds and amphibians on the basis of their characteristics.

Do you know some vertebrates are cold-blooded animals and some are warm-blooded animals? **What are cold-blooded and warm-blooded animals?** Are their body structures same or different?

Fishes, amphibians and reptiles are cold-blooded animals because they take on the temperature of their surroundings.

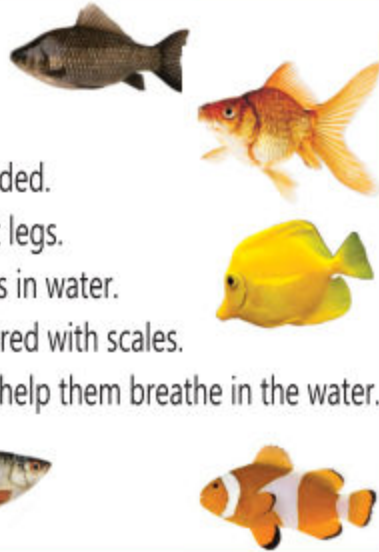
Birds and mammals are warm-blooded animals because they maintain a constant body temperature. This means that their body temperature does not depend upon the temperature of their environment.



Classification of vertebrates

Fish

Fish live in water.
Fish are cold-blooded.
They have fins not legs.
They lay their eggs in water.
Their body is covered with scales.
They have gills to help them breathe in the water.



Amphibians

Amphibians live on land and in water.
Amphibians are cold blooded.
They have moist skin.
They lay their eggs in water.
(Frogs, toads, newts and salamanders are amphibians.)



Reptiles

Reptiles have dry scaly skin.
They are cold-blooded.
They lay their eggs on land.
Reptiles live in water and on land.





Figure 1.11: Classification of vertebrates

Activity 4: Identify the structural differences among vertebrates. Select any one vertebrate and become an expert on it. Research your vertebrate and complete the profile given below. Also, write all your information on a chart.

Common Name: _____

Scientific Name: _____

Found in/or: _____

Feeding habits: _____

Unusual characteristics: _____

1. Display the chart in your class.
2. Share the picture, along with the characteristics, with other students.
3. Record all the structural differences among vertebrates on your notebook.

Teacher Note: Explain this activity and instruct the students in doing the research work.

Invertebrates



Identify key characteristics of worms and insects.

Worms:



Do you think that earthworms are insects? Earthworms and insects are alike in many ways, but earthworms are not insects. **What are earthworms?**

Earthworms are worm- like invertebrates. Other examples of invertebrates are leech, flatworm, tapeworm and roundworm. Some worms have segmented bodies; others do not have segmented body. They do not have legs. Some of them suck blood like leech and roundworm. Some worms are even environment friendly like, the earthworm.

Activity 5: To find out the characteristics of different worms.
'Secret code for worm lovers'

What to do:

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

To decode the secret words, use the code below.

Earthworms can also be called 14 9 7 8 20 3 18 1 23 12 5 18 19.

A tapeworm lives inside the 8 21 13 1 14 body.

As earthworm 2 21 18 18 15 23 through the soil, they give plants the 1 9 18 that they need.

A leech sucks 2 12 15 15 4 through its suckers.

Teacher Note: Engage students in performing the activity in groups or demonstrate it. Arrive at a conclusion along with the students.

Insects:

Do you know the name of the world's heaviest insect? Its name is goliath beetle. It can weigh over 100 gm. It lives in Africa.

All insects have three pairs of jointed legs, and in addition to this, many of them have one or two pairs of wings. Some of them have an antennae on their heads. Their body is divided into three main segments. They suck our blood or nectar of flowers. And they lay eggs.

Activity 6: Find out the characteristics of different insects.

Word Bank

Lady bug	Mosquito
Cockroach	Dragonfly

What to do:

Use the following hints and the Word Bank to decide which insect each riddle describes.

1. I have two pair of long, thin wings.
I eat mosquitoes and other small insects.
I live near lakes, ponds, streams and rivers.
My abdomen is very long, as long as a darning needle.

What am I? _____.

2. I like warm, damp and dark places and come out at night.
Humans hate me.
I am a destructive household pest.
I am closely related to grasshopper and crickets.

What am I? _____.

Insect's world!

Have you ever wondered why there are so many insects?



There are many reasons why insects outnumber any other group of animals in a count of heads!



Because they are small, they need less food.



Their small size helps them to hide from enemies.



Classification and characteristics of plants

- 1 Compare flowering and non-flowering plants.
- 1 Classify the flowering plants into two major groups and give examples of each group.



Do you know? Scientists have discovered over 380,000 different species of plants.

Plants are different from animals in such a way that they are rooted to one place and they use spores or seeds for reproduction. They also make their own food using carbon-dioxide and water in the presence of sunlight, with the help of chlorophyll, which is in their leaves, in a process called photosynthesis.



Do you know what plants do to give us food?



What food did you eat today? Did anything in it come from plants?

Do you have any plants at your home? Can you identify which type of plant is it?

How many types of plants do you have at your home?
Which type of plant is more?

All the plants on the earth make up a kingdom, called the **plant kingdom**. Plants are divided into two main groups, those which have flowers are called flowering plants and others without flowers are called non-flowering plants.

Unscramble the given word:

TNAOBY

Hint: The scientific study of plants.

Classification of plants

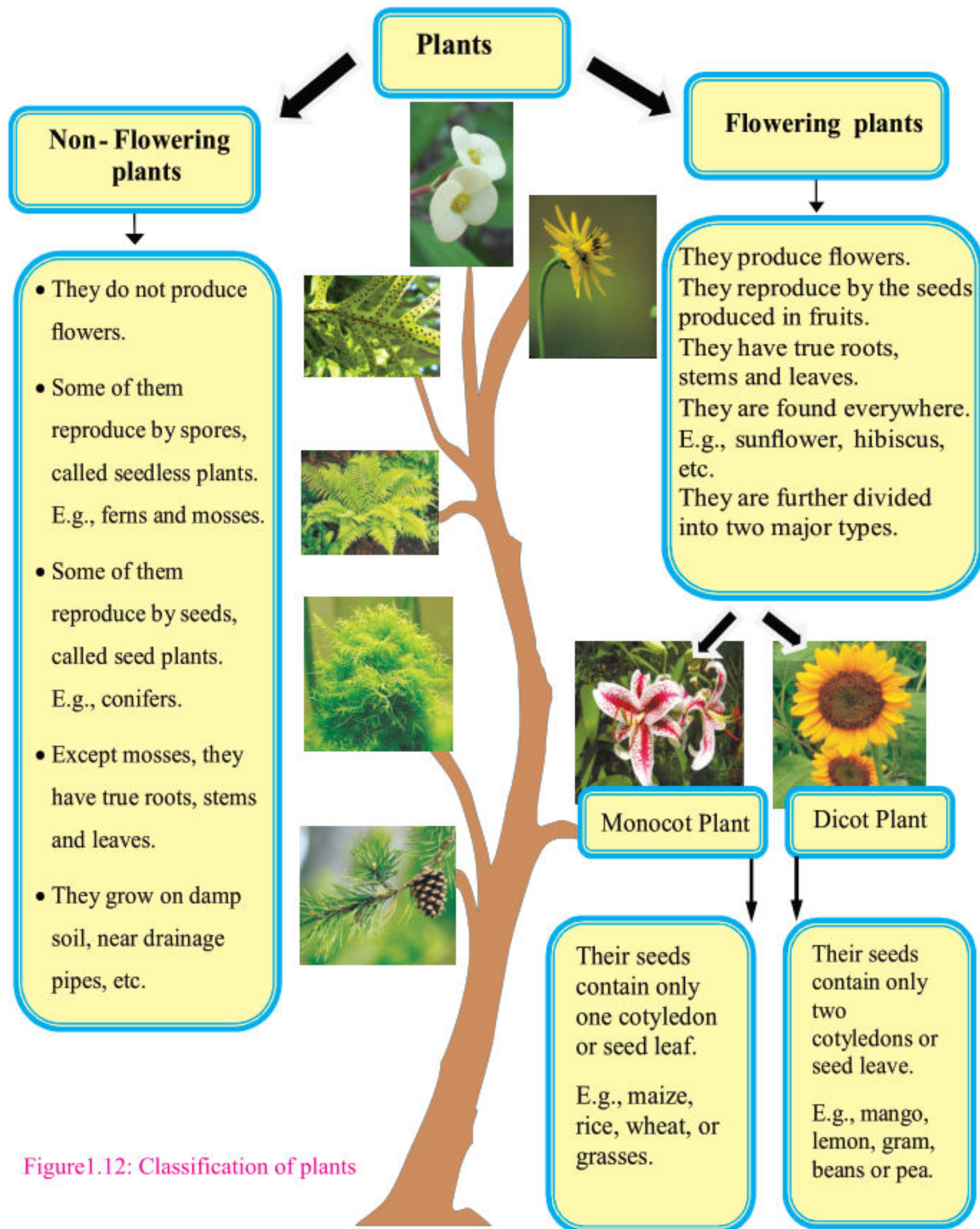


Figure 1.12: Classification of plants

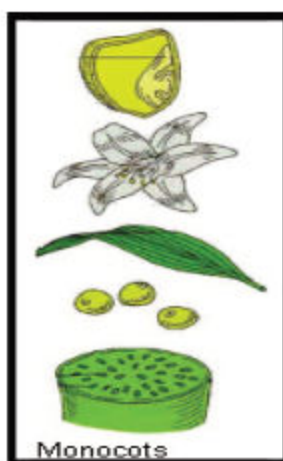
Characteristics of monocot and dicot plants

- Compare the structure of a monocot and a dicot seed.
- Compare the structure of a monocot and a dicot leaf in terms of its shape and venation.
- Differentiate between the structure of monocot and dicot flowers in terms of number of floral leaves.



Am I a monocot or a dicot plant?

Do you know the main differences between monocot and dicot plants?



Monocots

Monocotyledonous plants	Dicotyledonous plants
One cotyledon or seed leaf.	Two cotyledons or seed leaves.
Generally marked parallel leaf venation.	Generally marked netted venation of leaves.
Flower parts typically in groups of three or multiples.	Flower parts typically in groups of four or five.



Dicots

Activity 7: Identify the structural differences between monocot and dicot plants.

Name the following plants. In the circle provided, write 'M' if it is a monocot and mark a 'D' if it is a dicot.



Teacher Note: Engage the students in performing the activity in groups. Also assist the students in identifying the characteristics of monocot and dicot plants

Activity 8:

What I need:

- Visit a nearby nursery or farm with your class, where you can see a variety of plants. Take your notebook, pencil, and the pictures of seeds, leaves and flowers of both monocot and dicot plants with you.



Figure 1.13: A nursery

- Ask the nursery keeper/farmer to give you the six different types of seeds of both monocot and dicot plants.
- Soak these seeds in a container. Draw all the types of seeds and also write their names. Now slowly peel the outer covering of the seed. Observe how many cotyledons it has. You may need to use the hand lens for cotyledons.
- Observe different plants and note down the differences in their leaves and flowers.
- Collect six different types of leaves and flowers of both monocot and dicot plants.
- Examine each leaf carefully for its characteristics like shape and venation. Sketch a drawing of each leaf on your notebook. Discuss your findings with your pair and teachers. Also compare your sketches with the sample picture.
- Examine each flower carefully for its characteristics, like the number of floral leaves. Sketch a drawing of each flower on your notebook. Discuss your findings with your pair and teachers. Also compare your sketches with the sample picture.

What to do:

1. Your teacher will form groups of six students in a team.
2. Ask each member to observe and record characteristics of one leaf, one flower and one seed from the nursery.
3. Draw or bring a picture of the monocot and dicot plants.
4. Along with the teacher, observe the monocot and dicot plants, and their characteristics.
5. Now share the name/picture of the monocot and dicot plants, along with the characteristics, with group members.

Teacher Note: Discuss with students and assist them in identifying and recording the characteristics of monocot and dicot plants. Also, bring pictures of monocot and dicot plants for observation.

What I observed: (Remember to write your team member's names.)

Team members	Name/Picture of monocot and dicot plants	Characteristics of Monocot and Dicot plants (use tick)					
		Seeds		Leaves		Flowers	
		One cotyledon	Two cotyledons	Parallel venation	Netted venation	Floral leaves three or multiple of three	Floral leaves four or multiple of four

Activity questions:

1. What characteristics were most common in monocot and dicot plants?
2. What characteristics were different in monocot and dicot plants?

Teacher Note: Engage the students in the activity and help them to identify similarities and differences in monocot and dicot plants

Summary

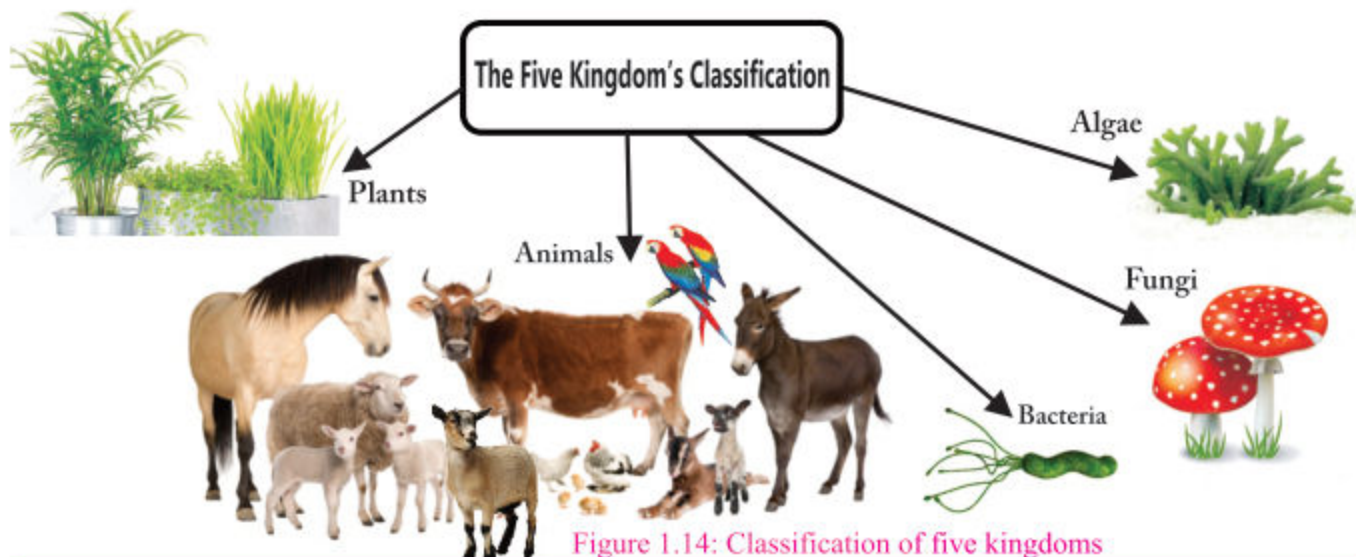
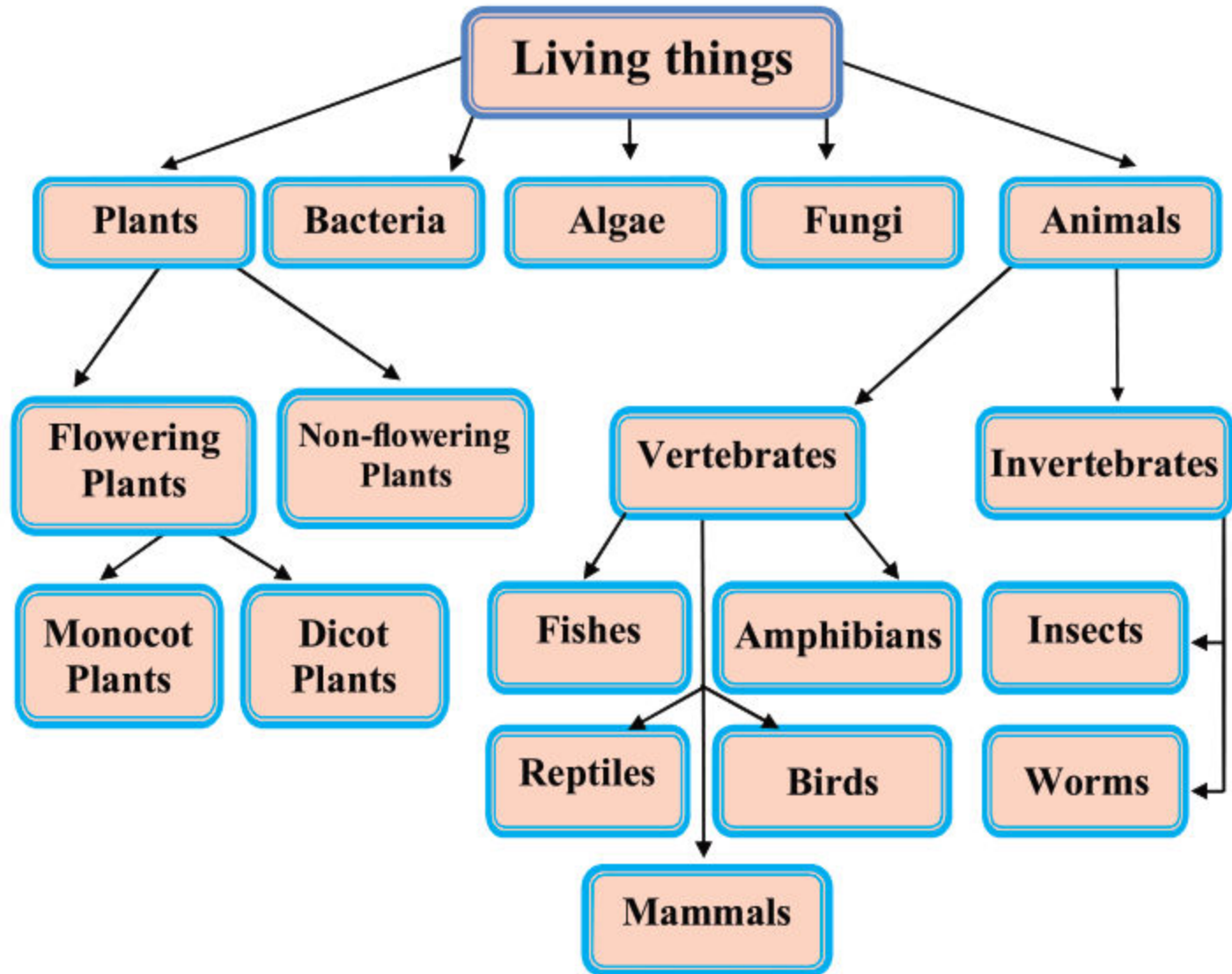


Figure 1.14: Classification of five kingdoms

Review questions :

1. Circle **T** for True and **F** for False Statements:

- | | | |
|--|---|---|
| a) spider is an insect. | T | F |
| b) mushrooms belong to plant kingdom. | T | F |
| c) leech is an example of an invertebrate. | T | F |
| d) dicot leaves have parallel venation. | T | F |
| e) all insects have two pairs of wings. | T | F |

2. Circle the best answer.

- i. Which one of the following is an example of amphibian?
 - a) lizard
 - b) salamander
 - c) turtle
 - d) bat
- ii. Which one of the following is not an example of mammals?
 - a) cat
 - b) bear
 - c) newt
 - d) dolphin
- iii. The body of birds is covered with:
 - a) scales.
 - b) hair.
 - c) feathers.
 - d) fur.

- iv. Which of the following is not a characteristic of a monocot plants?
- a) Its leaves have parallel venation.
 - b) Its seeds consist of one cotyledon.
 - c) Its flowers have three or multiple of three floral leaves.
 - d) Its flowers have four or multiple of four floral leaves.
- v. Which one of the following incorrectly shows the types of vertebrates through its examples?

	Types of vertebrates	Examples
a)	Reptile	Snake
b)	Mammal	Shark
c)	Bird	Ostrich
d)	Amphibian	Toad

3. Give brief answer to the following questions.

- i. What is meant by classification? Why do scientists classify things?
- ii. Define the following terms with examples of each:
 - a) insects b) worms
- iii. Think of two different kinds of vertebrates. List at least two similarities and two differences between them.
- iv. How would a scientist describe the following:
 - a) a fish b) a mammal c) a non-flowering plant
- v. A crocodile can also spend some time in water. List at least three ways in which it is different from a frog.

4. Flowering plants comparisons

Use the Venn diagram below to compare the following plants. Fill in the circle below the monocot plant with characteristics common only to the monocot plant. Fill in the circle below the dicot plant with characteristics common only to the dicot plant. Where the circles overlap, fill in characteristics both flowering plants share. Write a story about your findings on another piece of paper.



Monocot plant

Similarities

Dicot plant

